

✂ HOW SATELLITE DATA WORKS - CRYSTAL CLEAR

The Big Misunderstanding

text

What you think happens:

Your Laptop → Talks to Satellite → Gets Data
(Like WiFi to satellite)

What ACTUALLY happens:

Satellite → Takes Photo → Sends to ISRO/ESA
→ They store it on their servers
→ They make it FREE to download
→ You download it like any file
→ Your AI reads that file
Simple Real World Analogy
text

Think of it like YOUTUBE:

Satellite = Camera crew filming everywhere
ISRO/ESA servers = YouTube servers
Satellite images = YouTube videos
You downloading images = You watching videos

YouTube camera crew does NOT
connect to your laptop directly

They film → Upload → You download → You watch

SAME WITH SATELLITES:

Satellite films Earth → ISRO uploads images
→ You download → Your AI reads them

Your AI never talks to any satellite
Your AI reads DOWNLOADED FILES
that satellites already captured
Where The Free Data Comes From
text

THREE MAIN FREE SOURCES:

SOURCE 1: European Space Agency (ESA)

Website: scihub.copernicus.eu
Satellites: Sentinel-1 (SAR) + Sentinel-2 (Optical)
Coverage: Entire Earth
Update: Every 5 days for same location
Cost: COMPLETELY FREE
How: Create account → Search location → Download

SOURCE 2: ISRO Bhuvan

Website: bhuvan.nrsc.gov.in
Satellites: Cartosat, RISAT (Indian satellites)
Coverage: Entire India
Cost: FREE for research
How: Register → Request data → Download

SOURCE 3: NASA

Website: earthdata.nasa.gov

Satellites: Landsat, MODIS
Coverage: Entire Earth
Update: Daily to weekly
Cost: COMPLETELY FREE
How: Create account → Search → Download
What Actually Happens Step by Step
text

STEP 1: Satellite Takes Photo

Sentinel-2 satellite flies over Bihar
Takes high resolution photo
Sends photo to ground station in Europe

STEP 2: ESA Stores the Photo

ESA processes the raw photo
Converts to standard format (GeoTIFF)
Uploads to their free public server
Anyone can download it

STEP 3: YOU Download the Photo

You go to scihub.copernicus.eu
Search: "Bihar, India, June 2024"
Click download
File comes to your computer
(Like downloading any image from internet)

STEP 4: YOUR AI Reads the File

Your AI opens that downloaded file
Processes it locally on your laptop
Answers questions about it
No internet needed after download
No satellite connection needed
Ever.
For The SIH Demo Specifically
text

HOW YOUR DEMO WILL WORK:

BEFORE THE HACKATHON:

You download sample images
from ESA + NASA + ISRO Bhuvan
Store them on your laptop/server
These are your demo images

DURING THE DEMO:

Judge sits in front of your screen
You upload a pre-downloaded satellite image
Judge types: "What do you see here?"
Your AI reads the local file
Gives answer in 10 seconds
No internet needed at all

LIVE QUERY OPTION:

You can also connect to ESA API
User types location: "Mumbai"
Your system automatically downloads
latest satellite image of Mumbai
Then your AI analyzes it
This takes 30-60 seconds but

feels like magic to judges
The Two Modes Your System Has
text

MODE 1: Pre-loaded Images (Safe Demo Mode)

You pre-download 20-30 satellite images
Store them locally
Demo uses these stored images
Advantage: Fast, reliable, no internet needed
Use this: For hackathon demo

MODE 2: Live Download (Impressive Mode)

User types any location on Earth
Your system connects to ESA server
Downloads latest satellite image
AI analyzes it live
Advantage: Works for ANY location
Use this: If internet available at venue
BigEarthNet Dataset (For Training)
text

The PS says use BigEarthNet for training

What is BigEarthNet?

It is a collection of 590,000
satellite image patches already
downloaded and packaged for you

Someone already:

- Downloaded from ESA satellites
- Labeled each image
- Packaged everything neatly
- Made it free to download

You just download BigEarthNet
= You have 590,000 satellite images
= Ready to train your AI
= No satellite communication needed
= Just like downloading a dataset
from Kaggle or HuggingFace

Size: ~65GB total
Download: bigearth.net
Time: Few hours on good internet
Complete Picture in One Diagram
text

SATELLITES (Flying in space)

↓

Take photos of Earth

↓

GROUND STATIONS (In Europe/India/USA)

↓

Receive and process photos

↓

FREE PUBLIC SERVERS (ESA/ISRO/NASA)

↓

Store millions of images

↓

YOU (Sitting in Pune)

↓

Download images like any file
↓
YOUR AI (Running on laptop/GCP)
↓
Reads downloaded files
↓
Answers questions about them
↓
JUDGES (Impressed)

At NO POINT does your AI
talk to any satellite directly

One More Simple Example
text

Think of Google Street View:

Google car drives everywhere
Takes millions of photos
Uploads to Google servers
You view them on your phone

Does your phone talk to Google's car?
NO.
Your phone downloads stored photos.

SAME THING:

Satellite flies everywhere
Takes millions of photos
ESA uploads to their servers
Your AI downloads and reads them

Your AI never talks to the satellite.
Your AI reads stored photo files.
That's it.
So What Is Your Project Really?
text

A very smart image reading system
that understands satellite photos
and answers questions about them
in plain English

The satellite part already happened
Someone else already took the photos
They gave them to you for free
You just need to make AI that
reads and understands those photos

That is your entire project.
Reading satellite photos intelligently.
Nothing more. Nothing less.